Indian Institute of Technology Kanpur

Department of Mathematics and Statistics

Introduction to Bayesian Analysis (MTH422A)

Assignment 2, Due date: February 29, 2024, Thursday

Answers should be provided neatly. All rough work should be attached with the original submission, along with the original submission but not written together. Commented code should be provided as a separate .R file. If rough work is not shown and you cannot justify that the final answer can be thought very straightforwardly, you will be penalized for cheating. In case of cheating, all students involved will get zero. For `R` codes, please submit the Markdown file (.Rmd) as well as its PDF output. Thus, there should be a total of four/five files to be submitted-one PDF with neat solutions, one PDF with rough work (if necessary), one .R file containing the rough codes, one .Rmd file, and one .pdf which is the output of the .Rmd file.

1. Suppose $X_1, \dots, X_n | \mu, \sigma^2 \overset{IID}{\sim} \mathrm{N}(\mu, \sigma^2)$, with prior choice $\pi(\mu, \sigma^2) = \pi(\mu|\sigma^2)\pi(\sigma^2)$ with $\sigma^2 \sim \mathrm{IG}(a, b)$ and $\mu|\sigma^2 \sim \mathrm{N}(0, c\sigma^2)$, for large c and small a, b. Calculate the marginal posterior distribution $\pi(\mu|X_1, \dots, X_n)$. Calculate the posterior predictive distribution of X_{n+1}, i.e., $\pi(X_{n+1}|X_1, \dots, X_n)$ analytically. (1+1=2 points)

2. Bayesian two-sample T-test: Suppose $X_1, \dots, X_m | \mu_1, \sigma^2 \overset{IID}{\sim} \mathrm{N}(\mu_1, \sigma^2)$ and $Y_1, \dots, Y_n | \mu_2, \sigma^2 \overset{IID}{\sim} \mathrm{N}(\mu_2, \sigma^2)$, with prior choice $\pi(\mu_1, \mu_2, \sigma^2) = \pi(\mu_1|\sigma^2)\pi(\mu_2|\sigma^2)\pi(\sigma^2)$ with $\sigma^2 \sim \mathrm{IG}(a, b)$ and $\mu_1, \mu_2|\sigma^2 \overset{IID}{\sim} \mathrm{N}(0, c\sigma^2)$, for large c and small a, b. Calculate $\pi(\mu_1 - \mu_2|X_1, \dots, X_n)$. Compare your analytical result with a numerical one (both Monte Carlo or numerical integration are fine); for data simulation, select true parameter values $\mu_1 = 1, \mu_2 = 1.5, \sigma = 2$ and sample sizes $m = 25, n = 30$. (1+1 = 2 points)

3. Suppose $X_1, \dots, X_m | \lambda_1 \overset{IID}{\sim} \text{Poisson}(\lambda_1)$ and $Y_1, \dots, Y_n | \lambda_2 \overset{IID}{\sim} \text{Poisson}(\lambda_2)$. We assume the priors $\lambda_1, \lambda_2 \overset{IID}{\sim} \text{Gamma}(a, b)$. Calculate the posterior distribution of $\theta = \frac{\lambda_1}{\lambda_1 + \lambda_2}$. Further, simulate data for true values of the parameters $\lambda_1 = 2$ and $\lambda_2 = 2.5$, and the sample sizes $m = 10$ and $n = 15$. Calculate the 95% HPD credible interval of θ and comment on how you draw conclusion about the null hypothesis $H_0 : \lambda_1 = \lambda_2$ versus the alternative $H_A : \lambda_1 \neq \lambda_2$. (1+1=2 points)

4. Suppose $X_1 \sim \mathrm{N}(0, 1)$, and $X_{t+1}|X_t \sim \mathrm{N}(\rho X_t, 1 - \rho^2), t = 1, 2, \dots, T$. For the data generation, use $T = 10$ and $\rho = 0.5$. Suppose we choose the prior $\rho \sim \text{Uniform}(-1, 1)$. Use acceptance-rejection to draw 10^5 samples from the posterior distribution as optimally as possible and plot the kernel density estimate of $\pi(\rho|X_1, \dots, X_T)$. Provide a numerical estimate (using a heatmap) of the joint posterior predictive density $\pi(X_{T+1}, X_{T+2}|X_1, \dots, X_T)$. (1+1=2 points)

5. Suppose we want to draw samples from the random vector (X, Y) following the bivariate normal distribution with means $\mu_1 = 0$, $\mu_2 = 0$, and standard deviations $\sigma_1 = 1$, $\sigma_2 = 2$, and correlation $\rho = 0.5$. Draw $B = 10^5$ samples from this distribution using the function `rmvnorm` and draw the scatter plot with 2D-kernel density heatmap. Now, suppose we want to draw samples in a Markov fashion $\{(X^{(b)}, Y^{(b)}), b = 1, \dots, B\}$. Let us assume the initial samples be $X^{(0)} = 0, Y^{(0)} = 0$. Then draw $X^{(b)}$ from conditional distribution of X given $Y = Y^{(b-1)}$, and draw $Y^{(b)}$ from conditional distribution of Y given $X = X^{(b)}$ for $b = 1, \dots, B_0 + B$ in an iterative fashion, where $B_0 = 10^3$. Discard the first B_0 samples and draw the scatter plot with 2D-kernel density heatmap based on the remaining B samples. Comment on the differences in the pattern observed in the two heatmaps. Compare the histograms or the 1-D kernel density estimates of the samples individually for X and Y separately and comment on similarities and differences you observe between these two sampling schemes. (1+1 = 2 points)